

Model-wise thinking

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Part 1: Introduction to model-wise thinking; reasons to be interested in it

Introduction

- This presentation is about thinking in terms of models.
- Some other terms for potentially the same thing:
 - conceptions, representations, pictures, theories, ontologies
- This is a huge topic.
- This presentation is an opinionated introduction.

Examples of models

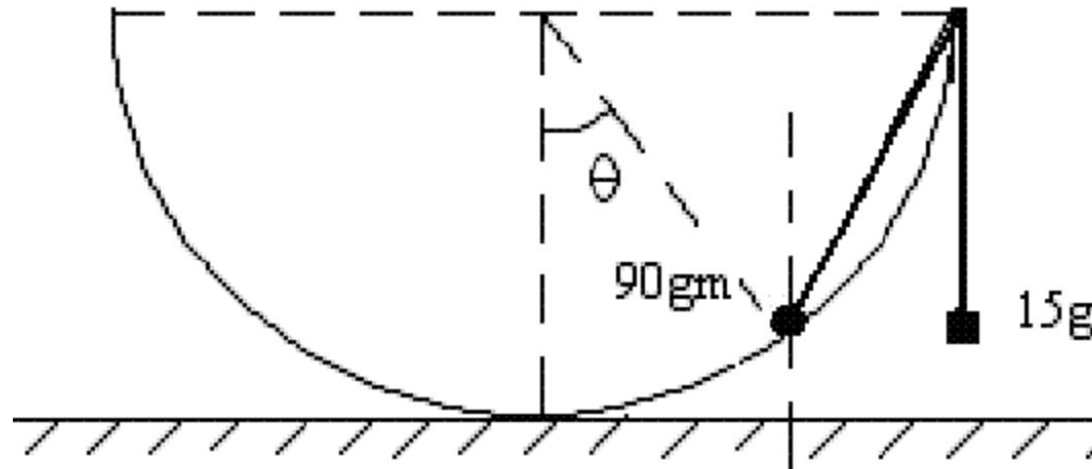
- I could have a classical-mechanical view of the situation in a physics problem.
 - I take there to be certain physical objects. I take each one to have certain attributes like its position, mass, acceleration, summary force acting on it, momentum, axis of revolution, moment of inertia around that axis, angular velocity and momentum, kinetic energy, etc..
 - I take there to be various forces acting on each object. Each force can also have a causing entity (but doesn't strictly have to I guess — e.g. the causing entity of gravity is often not in the picture). I take there to be stresses in ropes.
 - I take there to be axioms/principles governing relations between these various quantities. For instance, each force should have an equal and opposite reaction force. $F=ma$. The force between two surfaces is normal plus there can be a friction force of size at most μF , where F is the normal force. Strings have constant tension usually. Energy is conserved.
- Classical mechanics is itself also some sort of model, or maybe a sort of template/arsenal from which specific models can be constructed.

A specific example of a physics problem

A hemispherical smooth bowl is held fixed with its axis vertical. A particle of mass 90g attached to one end of a light inextensible string is placed inside the bowl. The string passes over the frictionless rim of the bowl and is attached to another 15g mass hanging freely in air at its other end. In equilibrium position of the system of particles find the angle θ subtended by the radius through the 90g particle with the vertical.

solution sketch:

- write force balance on 15g mass
- gives tension in string
- now write force balance on 90g mass
 - with θ as a free variable
 - using facts:
 - normal force is normal
 - string tension is const
- specifically in the direction perp to the normal
- have 1 equation in 1 unknown (θ)
- solve

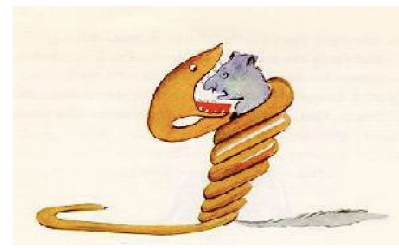


[another example](#)

Examples of models (pt 2)

- More broadly, any scientific theory is an example.
- You have an econ model of two countries, one of which has an absolute advantage over the other in the efficiency of producing any good. You prove a theorem saying that allowing trade would make both countries richer than autarky.
- You learn the rules of chess. Now, when playing, you know what the current position is and you know what moves are allowed. More specifically, you know what all the pieces are and how to tell which piece is which and what moves you're allowed to make with each piece, and you know each move has to involving moving with exactly one piece. You know what checkmate is.
 - When learning a new board game, it's a fun exercise to keep in track of how your model of the game is developing as the rules are being explained to you.
- As a somewhat better chess player, you might further track certain variables like the sum of the “heuristic values” of all your pieces (like, knights and bishops are worth 3 pawns, rooks are worth 5, queens are worth 9), piece development, whether you played f3 (never play f3), etc..

Examples of models (pt 3)



- Say you have a cube in front of you and you want to know how many rotational symmetries it has. You think about this by asking the same question of the abstract cube $[-1,1]^3$.
 - another fun example: Imagine hanging a cube from a vertex and slicing it horizontally into two equal halves. Now look at the 2D shape under the top half. What shape would you see?
- As you read the first paragraph of a novel, sentence by sentence, new structure/content is being added to your conception of the situation being described.
 - “Once when I was six years old I saw a magnificent picture in a book, called True Stories from Nature, about the primeval forest. It was a picture of a boa constrictor in the act of swallowing an animal. Here is a copy of the drawing.”
- Sometimes people say “here’s my model of X” — e.g. with X = [neural net learning] or [the conditions in which the development of a technology is rapid] or [political polarization in recent years] or [the broad suite of health effects of GLP] and then they communicate some sort of picture of X to you.
 - I think this communication usually happens by a combination of saying what variables to track, making some explicit claims, and displaying some reasoning (maybe this lets the other person gain your implicit inference rules?).
- When you learn arithmetic, maybe you gain some model of how the natural numbers work.

Examples of models (pt 4)

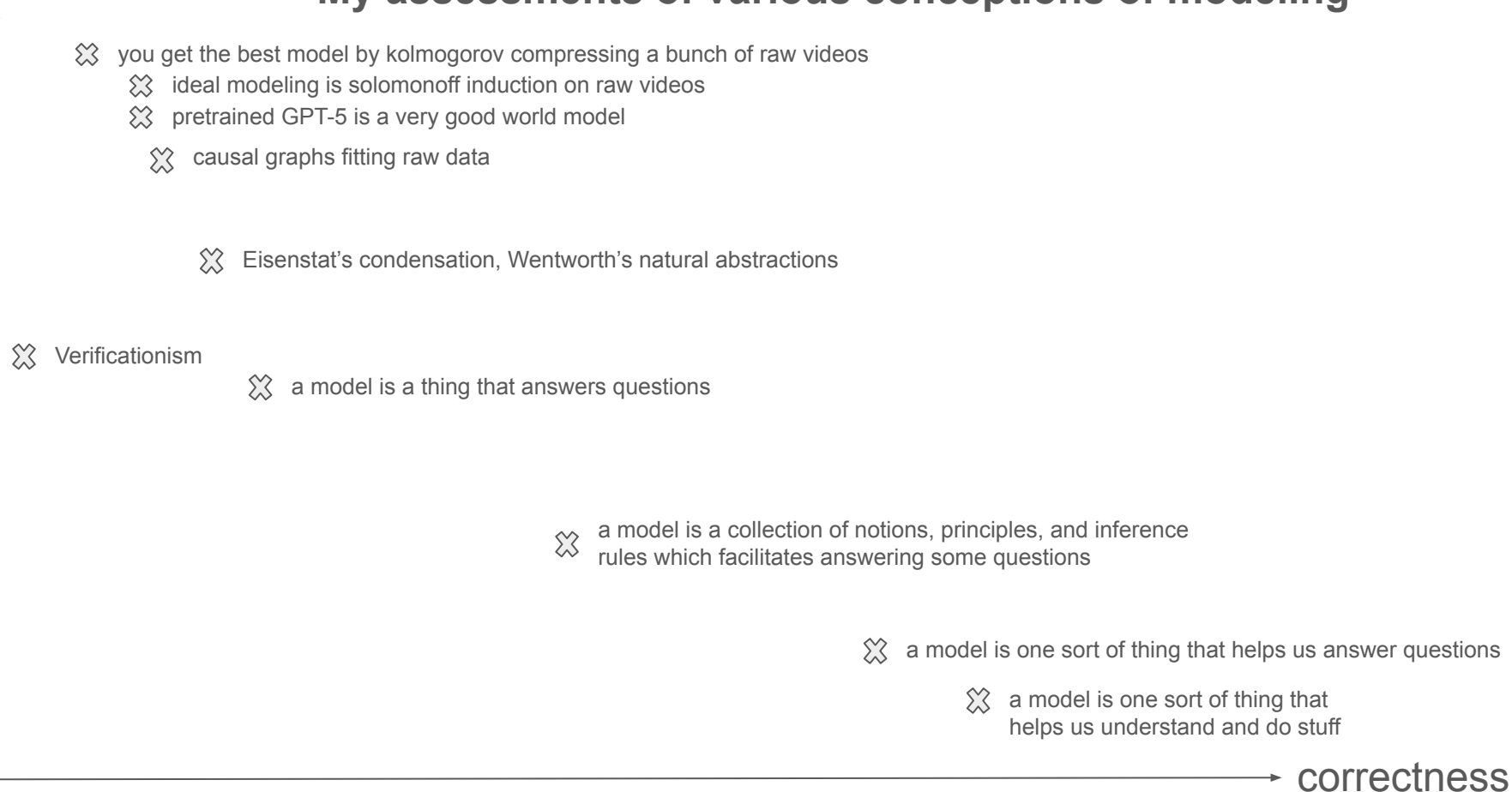
- When flipping a coin a bunch of times, you think of the coinflips as being independent 50/50 random variables.
- When running a medical trial in which some subjects get a candidate drug and others get placebo and you measure what happens to the condition of interest for each person, you might think of the i -th participant's condition as being an independent draw Y_i from one of two unknown distributions with unknown means (one distribution for those who took the drug, one for those who took a placebo).
- When trying to compute some number-theoretic quantity, sometimes one gets an “empirically” correct answer when thinking of which numbers are prime as being randomly determined with probability $1/(\log n)$ for the n -th number to be prime, sometimes with the further assumption that each prime has a uniform probability to have each nonzero remainder modulo smaller primes.
 - It is interesting to investigate the use of heuristic models specifically in mathematics. See Polya's Mathematics and Plausible Reasoning and ARC's work.
- Think about what you do when learning a dance from someone's demonstrations. This is not well-described as making a video of you have pixels which are close to a video of them. Instead, you are making what you do have the same key high-level features. This is perhaps revealing of what features are tracked in your conception of what they are doing.

Is there even a thing here?

- To me, it's an open question if all of these are examples of the same basic thing, of having a representation/conception/picture/model/theory of a situation.
 - Maybe there are multiple different things, or maybe there isn't really a thing at all. Maybe it's not a productive frame for understanding thinking.
- There's certainly a bunch of different stuff going on in having a model of a situation.
- Also, even if modeling is well-considered a separate phenomenon from the rest of a mind's operation, it will be heavily dependent on all the other stuff.
 - See Benson-Tilsen on [Indra's net](#).
- Maybe one would at least want to distinguish between proper well-defined scientific theories and loose mental models?
 - My position on this is that scientific models are much less well-defined than one might think. Newton's laws as a stand-alone object uncoupled from a human with "common sense" would be unable to solve basically any problems in a classical mechanics textbook.
- There is certainly a difference between situation-specific models and [model templates]/[conceptive arsenals] that one can construct specific models using. I'm going to be loose with this distinction.

concreteness

My assessments of various conceptions of modeling



Reason 1 to care about modeling: verification

- It would be useful for AI alignment if we could verify stuff which is not precisely mathematically specified.
- In arguments outside math, one thing that often shows up is model-thinking.
- So, it'd be useful to be able to tell when some model-thinking is trustworthy.
- So, maybe a better understanding of modeling would be useful for pushing principled verification forward.

Reason 2 to care about modeling: diamond maximizer, “(re)targeting a search”

Here’s a scheme for the diamond maximizer problem:

- Make a big data set of videos of the world.
- Have some sort of AI analyze these videos and get a good sense of the world. We wish to set things up so this AI gains a capability to think of the world in reasonable terms (like, it should have a diamond notion, or at least have an ability to understand what a diamond is), and to ask and answer lots of questions. In particular, we want it to be able to ask and answer “what sort of device should be placed in the world so that a lot of diamond gets made?”.
 - Why should this be possible? One reason is that it feels possible for us to do if we got a bunch of videos from a weird alien universe, at least with enough time and to some extent.
 - We can also let this AI interact with the world a bunch. Maybe it’s important to let it “ask the world its own questions”, in principle or just in practice.
- Try to get the AI to actually ask and answer this question, maybe with some mind editing.

If we understood what sort of thing a world model is, maybe we’d be in a better position to pull this off.

Reason 3 to care about modeling: getting good models

It'd be cool if we could ask an AI for good models of situations, so we could maybe use those to make useful technologies, answer important questions, etc..

Reason 4: interpretability, monitoring, ELK

- Maybe AIs do model-wise thinking.
- Maybe if we understood model-wise thinking better, we would be in a better position to understand an AI's thinking.
- This would be helpful for monitoring.
- Maybe it'd be helpful for making nicer AIs.

to give an analogy:

- It's good for interp that we have the idea that an AI could have beliefs, because that puts us in a better position to figure out whether it does and in what sense and what its beliefs are, which could enable e.g. better monitoring or adversarial training.

Reason 5: you use this understanding to make an understood nice AGI?

- Maybe you understand model-thinking and some other things and then you just write well-understood nice AGI code.

Reason 6: it'd be good to better understand how the following work: language, meaning, truth, problem-solving, learning, thinking

- The topic of model-thinking relates to many central philosophical and scientific questions about minds.
- So, it'd also just be really cool philosophically/scientifically to understand modeling better.

Some caveats on investigating modeling

- There doesn't have to be some single correct way to do model-wise thinking.
 - I think there is probably a very rich space of ways to do model-wise thinking.
 - There are probably also ways to think well without thinking in terms of models, at least if we mean something pretty precise by thinking in terms of models.
- It's probably natural for specific ways to do model-wise thinking to be rich and complicated.
 - My guess is that the way humans think in terms of models is probably really complicated.
 - Like, pulling some numbers out of my ass, maybe there are 10-100 ideas that are pretty much necessary for human-like modeling to get going, and 100-1000 more significant "compute multipliers".
 - Various ideas going into good model-wise thinking are invented by humans / cultural / gained over a lifetime. For example, the scientific method, the practice of trying to communicate your models to others
 - There's probably very many good ideas for doing model-wise thinking much better still to be invented.
 - I think it's plausible that the quantity of ideas for doing model-wise thinking that remain to be invented will always be greater than the quantity that have already been invented.
- I think we should be open to different ones of the aforementioned reasons to be interested in modeling to have basically different answers. E.g. there could be a kind of modeling one can use for writing verifiable physics olympiad solutions and a different kind actually implemented in an AI and useful for interpretability.

A brief literature review on modeling

- There is a large literature on modeling; I'm only familiar with a small fraction.
 - from philosophy: Kant, Wittgenstein, Quine, a bunch of philosophy of science, philosophy of mind, philosophy of language, specifically Montague semantics, logic, formal ontology, ontology
 - from linguistics: sentences being represented by syntax trees, semantics, [discourse representation theory](#)
 - from math: mathematical logic, category theory, a bunch of stuff in applied mathematics (specifically PDEs, dynamical systems, numerical analysis)
 - from GOFAI: Minsky's frames, Hofstadter's work on analogies, Pearl's work on causality, knowledge bases, specifically Lenat's [Cyc](#), [ontologies](#)
 - from cognitive psychology: work by [Johnson-Laird](#) and many others on mental models
 - from statistics and machine learning: statistical modeling, Mumford's Pattern Theory, arguably most of machine learning
 - from AI alignment: [Yudkowsky](#), [Soares](#), [Wentworth](#), [Eisenstat](#), [Benson-Tilsen](#), [Ngo](#)
 - from the rest of CS: scientific computing, data structures
 - a significant fraction of neuroscience is on what sorts of representations humans have of various things
- I'm probably missing various important things. Feel free to message me with those.

Hyperparameters of model-thinking

- I think it's better to see human thinking as constructing specific models for specific situations, and not as having a single unified world model.
 - Or if you want to see a grand human world model, then you have to think of this as being given in part in terms of the ability to create lots of small models for specific situations.
 - One reason for this is that different situations call for different contradictory assumptions (e.g. sometimes you think of the Earth as being flat; sometimes you think of it as being a ball, sometimes you think of it as being an oblate spheroid, sometimes you remember that there are mountain ranges and deep ocean floors, etc.).
 - That said, it is fair to subject different models to some coherence constraints, and this makes your many different models hang together somewhat.
- Ultimately, our models are for doing various things. E.g. for:
 - designing and building a clock
 - going to the Moon
 - winning in a game
 - not dying because of AGI
- An important special case: models are often constructed for answering questions.

Some major questions about model-thinking

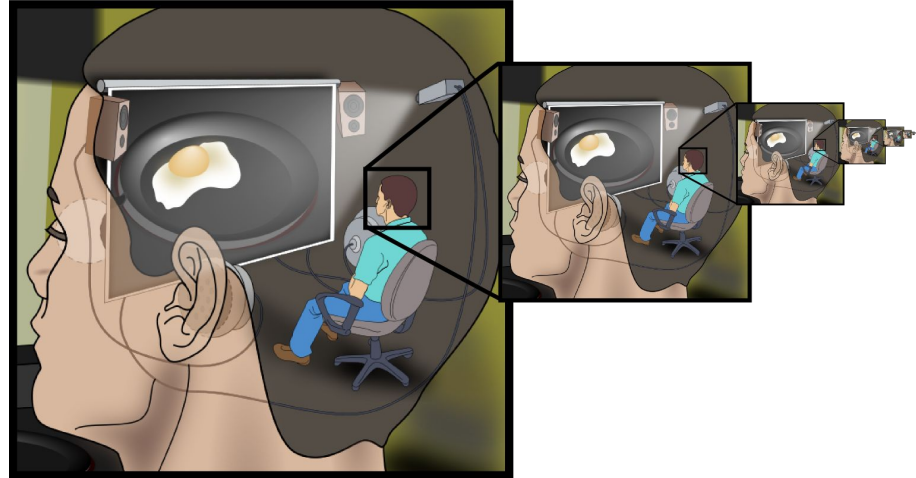
The rest of this presentation will discuss the following three big questions about model-thinking:

- What sort of thing is a model? What is the structure of a model? What is it to think in terms of a model? What sort of thing is model-wise thinking? (Is it even a thing?)
 - Often, model-thinking seems to involve many models. What are the structures of those? What is the structure of how they relate to each other?
- What is it for a model to be good? What's the criterion a model should satisfy? How does one “verify” or come to trust a model? Is there a straightforward reliable procedure for that?
- How does one invent/discover good models?

Part 2: What sort of thing is a model?

Homunculus views

- “Homunculus view” is a derogatory term in philosophy, for views that try to explain aspects of thinking by positing a smaller thinker “homunculus” looking at something.
- “How does vision work? Well, there’s like a screen inside your head, and your eyes are like a camera streaming video onto that screen, and then you’re like a guy looking at that screen.”



A homunculus view of modeling that actually isn't useless

- One sense of “model”: a physical object, usually smaller than the real object, that is used to represent something.
 - E.g. a model plane or a model bridge.
- Slightly more generally, this doesn't have to be a physical object.
 - E.g. you can make a digital model of an object for 3D printing or have a digital model of a city.
- Maybe model-wise thinking just replaces thinking of a real thing with thinking of a “small model” of it?
 - You have a map that is like the territory, and you think about the map.
 - One might also call this analogical/metaphorical thinking.
- This looks like a homunculus view of thinking (you've explained thinking of one thing in terms of thinking of another thing).
 - But I don't think it's trivial, because the latter thing can be much simpler/smaller.

Explicit examples of small models helping

- Your car is leaking some fluid, and you want to know if it is fine to keep driving for 50 km to a car repair. You pull up a schematic of the car to figure out what component is likely leaking, and then use that to make a decision.
- You use a map of the neighborhood to figure out how to go to a nearby store.
- You get a graph representation of town distances and you run a path-finding algorithm to give you the shortest path to a store.
- You might have encountered a type of logic puzzle where there are a bunch of constraints on some assignment of attributes to objects (like “neither Misty nor Rex is the German Shepherd”), with the goal being to deduce some specific attribute from the constraints (like who brought what dog to a dog show, and what breed each dog was). When solving such a puzzle, it is useful to maintain a grid representing known properties. These are called “[logic grid puzzles](#)”.

A more implicit example of the homunculus story in action

- Say we want to find the area of this room.
- We might go about this by measuring/guessing the two side lengths and then multiplying them together.
- How come we came up with this procedure and knew it would work?
 - If you had found the area of a rectangular room before, maybe you just had this as a memorized policy. But let's imagine you hadn't.
- I claim that you are relying on an analogy between the room and a rectangle.
 - There is some sort of “abstract object” (the rectangle) that you are somehow thinking about.
 - There are “bridging laws” between the abstract situation and then concrete situation.
 - You measure side lengths of the room and these become side lengths of the rectangle.
 - You compute the area of the rectangle and this becomes your guess for the area of the room.

Trying to make the homunculus view more precise

- So, the idea is that you are somehow thinking of a small/abstract version of a concrete situation. You're able to figure stuff out about this small version.
- You have “bridge laws” relating the concrete situation to the abstract situation.
 - E.g., having chosen a path on the map, you know how to walk it.
 - E.g., measuring the side lengths of this room gives you the side lengths of the rectangle.
- To give a more concrete thing:
 - There is a precise mathematical situation, with correspondences between concrete things and things in this mathematical situation.
 - For example, you have functions which are rabbit and fox populations and differential equations governing these functions, and then you can predict the lag time between their peaks or their maximal population sizes or whether they will go extinct.
 - For another example, you have a concretely implemented computer program, and you reason about what happens when it is run (e.g. you calculate its runtime) by reasoning about the abstract running of the abstract program.

A reason why any homunculus view is wrong

- Really, we don't just think about a model, we think in terms of a model.
 - If you force me to say something more specific, I'll say: a model somehow provides guidance for drawing inferences beyond just the guidance you get from considering an abstract object.
- Here's maybe the most obvious way to try to fix this:
 - Instead of having an analogy to a mathematical object, you have a theory in the sense of mathematical logic. To specify it, you say what the object symbols are, what the relation and function symbols are, what the axioms and inference rules are. Maybe also throw in some heuristics.
- For an example of a more interesting sort of “reasoning in terms of a model with given variables”, see [AlphaGeometry2](#).
 - In it, there is a part drawing cheap inferences one can prove with a given symbolic reasoning engine without introducing auxiliary geometric objects. We could think of this as giving the model of the system at a time.
 - And then there is an LLM-based part choosing which auxiliary points to add. We could think of this as expanding the model.

Rigidity issues with all the given models of modeling

- All models explicitly proposed so far in this talk are way too “rigid”.
- They will be missing lots of good inferences:
 - E.g. you forgot to put in your axiom base sth like “usually when a person is in New York, their left leg is in New York”, so you prove “John is in New York” but can’t prove “John’s left leg is in New York”.
 - In other words, “the models lack common sense”.
 - There intuitively needs to be some way to generate more correct inferences or axioms.
 - (Adding probabilities into the picture doesn’t fix this.)
- They get ridiculously confused when meanings or assumptions are slightly misspecified.
 - Maybe you think a dog has 4 legs, but then there’s a 3-legged dog in your situation. You observe a contradiction and conclude anything via principle of explosion. (Ok there’s a field of non-monotonic logics trying to address this but afaik they haven’t succeeded.)
 - If I’m doing interior design in a context where there are objects between that can be used as chairs or beds, maybe my model for doing interior design was written with an assumption that this does not happen and generates a contradiction.
 - In other words, these models aren’t robust.
 - In the human case, this sort of thing doesn’t happen. Somehow a model gets softly adapted to a situation?
 - It’s an interesting question if one could fix this by adding probabilities. My guess is that this can’t give a full fix because we do actually want some rigid meanings in a context, we just want these to be adaptive to the needs of the context.
- Any given bridge laws would be brittle. It seems like good bridge laws require the context of a lot of understanding.
- They don’t natively support adding new variables correctly.
 - E.g. maybe you had a rectangle class object and you forgot to implement that it has a “length of diagonal” variable, and so the model can’t handle that when you have to measure a diagonal instead of a side because the side is blocked.

Part 3: What is it for a model to be good?

The canonical criterion on a model: predicting correctly

- Yes, it is good for a model to make more correct predictions.
- However, not everything that makes correct predictions is a good model.
 - For example, if you ask the set of all programs for an accurate predictor of your video data set, this will be some cursed program that no one could use for anything other than predicting that data set.
 - Your superforecaster friend might make good predictions, but isn't a model. Probably a lot of programs inside Solomonoff induction are somewhat more like clever forecasters than like canonical models.
- Striving for concreteness, people say theories should predict raw data. But it is extremely rare for a scientific model to centrally predict raw data. Scientific models can predict stuff, or more broadly/correctly they can answer questions or solve problems, but these are almost always quite distant from raw data.
 - E.g. things like “will this cannonball hit the castle if we shoot it from this position at this angle?”, “how often does one need to reset a clock with this design because of accumulated errors?”, “at what temperature does ice melt at 120 kPa pressure?”, “how much fuel do we need to put in the rocket for it to get to orbit?”, “what is the age difference between twins after one has spent 10 years flying away from Earth at 0.7 c and then 10 years flying back again at 0.7 c?”.
 - You can make a scientific theory predict raw data by constructing sensors measuring these high-level things — then your theory predicts some pixels in a video of the sensor, okay. I think this is clearly a contrived way to turn high-level things into raw sensor predictions and you should just admit that your theory primarily answers high-level questions. The reason you think the sensor would be well-predicted is that you think your theory answers high-level questions.
- Most scientific models are not able to make almost any of the ordinary predictions they could be taken to make as stand-alone things. They are only able to make these predictions when skillfully wielded by a human scientist.
 - Or rather, a human scientist is able to answer questions by thinking in terms of a theory or model.
- Scientific theories also only predict stuff in very specific circumstances. Calculating most quantities in most situations involving messy rigid bodies is not tractable — this is not how physics problems are made.
 - We basically notice some regularity and construct a bunch of scenarios which can be understood because of that regularity.
 - For an extreme example, consider the theory of electric circuits; >99.99% of its applications are on very specific systems we've constructed. Optics too.
 - This is another major issue with the view that a model should predict raw data.

More criteria on a model

- A model should be true-ish.
 - For example, when you write a program and run it and think about what happens in practice by thinking about what happens when running that abstract program (e.g. to calculate its runtime), you kinda just know that your model is true-ish. Like, the abstract program which you think is being run is really being run.
 - If you propose a model of an atom where there is a dense central part carrying most of the mass around which there is a much less dense cloud of electrons, then it's good for your model if this is really true.
 - If you propose a theory in which angry gremlins are causing most storms, then there should really be gremlins who are angry and who cause most storms.
 - However, a model does not literally have to be true. E.g. it's fine to think of the Earth as being flat in many contexts, even though it is false that the Earth is flat. [When modeling ocean waves, people often assume that the ocean is infinitely deep](#), knowing very well that it really isn't. Intuitively, it's fine for an “axiom” of a model to be false as long as it is “true enough for the purpose at hand”. I'd like to better understand what that means. Maybe it can be cashed out in terms of saying there is a layer of claims downstream of the assumption that are nearly true? (E.g. maybe you use an assumption to derive some equations between forces where each term is approximately correct.)

More criteria on a model (pt 2)

- More on how a model should be true-ish:
 - There's a kind of philosopher who is upset at me here, saying something like "truth should not play a role in an explanation of modeling, because we would like to explain truth in terms of modeling, and then our explanation of modeling would be circular".
 - My response to this is that it would indeed be bad to give a genuinely circular picture of modeling, but I think it is fine to give a recursive picture of modeling where judging a model can involve further model-making and model-judging, perhaps in principle arbitrarily many times.
 - Or really I should give a recursive account of answering questions (or even figuring out how to act), which involves modeling and other things.
 - The reason for this is that judging a model doesn't make calls to more modeling, but instead in general to asking more questions.
 - This is like how you can have a list sorting algorithm that calls itself on the first half of the list and on the second half of the list and then merges the two ordered lists into one ordered list. Okay, here you need there to be base cases, namely to do nothing on a list with at most one element. In the case of answering questions, my guess is that there would not fundamentally be base cases, the recursive questioning would just stop somewhere, maybe where various potentially questionable things appear intuitively clear.
 - There's also a philosopher who complains "truth is some woo-woo nonsense, only raw data predictions are real". My response to this is: "no, and look at the anti-verificationist literature for more".

More criteria on a model (pt 3)

- A model should be simple, understandable, and easy to work with (e.g. it's good for computations in/using the model to be quick).
- “meta-induction”: A model should conform to inductive constraints on models.
 - For example, when trying to understand a situation involving solid objects interacting with each other non-electromagnetically, think in terms of newtonian mechanics (their positions, masses, accelerations, forces (gravity, normal forces, friction), momenta, moments of inertia, angular momenta, energies).
 - For example, models that view ordinary inanimate things and processes as having certain causally involved telic/value-like aspects currently have a bad reputation. (E.g. people used to think that heavy objects want to move closer to the center of the earth. People used to think that metals are striving toward being gold.)
 - You could think horseshoe theories have a bad reputation.
 - I'm not sure if this should be seen as an independent property, or just something that goes into assessing whether the model is true.
- More generally, you can just think about whether a model is going to help you solve problems, however you think.
- A model should help you understand things you want to understand. It should explain a priori confusing things. Explaining more different things is better.
- A model should help you create useful technologies and plan and act.
 - I think this might ultimately be the only criterion on our models as humans, with all the aforementioned criteria being instrumental to this.

Some concluding thoughts on judging models

- I haven't given a concrete criterion on models, a concrete procedure for judging models, a loss function on models.
 - A good procedure for judging models might be pretty different in the case of models in a mind vs models presented publicly e.g. when making an argument.
- I think it's a very interesting open question to find an effective procedure for judging if a model is good/valid/correct.
 - It's not clear there is such a procedure to be found. Maybe it turns out that evaluating models is more of an "art". That would be sad for verification hopes. But I have some hope.
 - One option is that efficient verification is kind of possible but it's defeasible and not robust, so can't be used in adversarial situations. That'd be sad for the verification hope.
- My best guess for a more concrete open problem here is to try to make a principled verification system for physics olympiad problem solutions (including a system for writing these solutions).
 - Maybe identify some smaller easier class inside physics problems that is still not just math problems, create a system for that class first, then try to expand from there.

Part 4: How does one construct good models?

Constructing good models

- There cannot be an efficient procedure for coming up with good models in general.
 - It is NP-hard to come up with a simple model or to check if there is a simple model.
 - [There are variants of the minimum circuit size problem which are known to be NP-hard.](#)
 - Also, bayesian inference with circuits with given structure and some free parameters is NP-hard.
 - There is no algorithm that takes any bitstring as its input and outputs the shortest program which outputs that bitstring. That is, Kolmogorov “best compressions” are uncomputable.
 - Proof: Suppose there is such a program P, call its length n . Then we can write down a program Q of length at most $100n + 10000$ which calls P on all strings in order, looking for the first one whose compression is longer than $100n + 10000$, and outputting that one. But then Q is a shorter compression of that string, a contradiction.
 - This is analogous to the situation for proofs: telling if a statement has a short proof is NP-hard, and [saying if there is a proof at all is undecidable](#).
- As a baseline, If you had a way of telling if a model is good, I guess you could just construct a bunch of models and then check if each one is good and keep the good ones. However, that’s very slow.
- My guess is that constructing good models is an “art”, and not something there is a “procedure” for.
 - I think it’s like how constructing proofs is an art (even though there is a procedure for checking proofs).
- However, given that humans can construct good models to some extent, there are practical procedures that construct good models at least that well.
 - There are probably systems that come up with good models much much more compute-efficiently than humans.
- Even if coming up with good models is an art, that doesn’t mean there’s nothing interesting to be said about it.
 - If anything, it means that there is a huge amount of interesting stuff to be said about it. Nearly all good ideas for coming up with good models probably haven’t been invented/discovered+implemented yet.

Models are made of parts

- When solving a physics problem, you construct a model involving some familiar notions: bodies, velocities, forces, fields, energies, pressures etc..
 - These can come with familiar laws/principles/axioms/[inference rules] attached.
- It seems like your model is basically composed of these parts, or maybe of tokens of given types.
- In addition to concepts being helpful for coming up with models, models made of familiar concepts are much easier to understand.
 - Consider learning the rules of a board game. Note that to someone who only understands math, we would define the same board game without words evoking common concepts like “road” and “action point” and “victory point” and “attack”. But this makes it much easier for a human to understand the rules of a board game.
- Three questions prompted by this:
 - How does one compose concepts into a model?
 - How does one come up with which concepts to compose?
 - How does one invent/discover good concepts to add to one’s arsenal?
- In the remainder of the presentation, I’ll consider the last question.
 - That question is also relevant to “what makes a model good?”, because a model being good is probably caused in significant part by being made of good concepts.

What concepts are good to have?

- My best guess for an ultimately correct criterion on good concepts:
 - Good concepts are ones that facilitate making good models.
 - You find new good concepts by trying to solve problems, sometimes somehow coming up with a new concept that lets you solve a problem, and then keeping it in your arsenal. You then use it in other contexts.
- But one can hope for some more straightforward/objective “heuristic criterion”.
- Here’s my best guess for that:
 - Good concepts are ones that make useful inferences easy.
 - These inferences could be predictions:
 - “If it rains and I go out, I’ll get wet.”
 - Or they could be more directly action-guiding:
 - “If it rains, I shouldn’t go out, or at least I should take an umbrella if I go out.”
- I will explain this hypothesis in more detail for mathematical concepts.

open problem: Give a criterion on mathematical concepts

- Concepts are somewhat less confusing in math than in general, because we can concern ourselves with objects that have precise definitions.
 - That is, we can bypass a good chunk of our confusion about the nature/[type signature] of concepts.
- A lot of the interesting stuff is still captured by the following question:
 - What mathematical concepts should one have?

The inferentialist hypothesis in math

- Consider the concept of continuity for a function.
- If a function is continuous, then it:
 - has a maximum on any closed interval
 - satisfies the intermediate value property
 - has a well-defined integral
- There are also lots of nice properties implying that you have a continuous function:
 - polynomials and exponentials and trigonometric functions are continuous
 - a composition of continuous functions is continuous
 - products and linear combinations of continuous functions are continuous
 - the uniform limit of a sequence of continuous functions is continuous
 - the integral of a continuous function is continuous
- The fact that there are lots of inferences from other properties to continuity and from continuity to other properties means that having the notion of continuity in your conceptual vocabulary opens up a lot of easy inferences for you.
 - The hypothesis says that continuity is a good concept largely by virtue of this.

The inferentialist hypothesis in math (pt 2)

- Motivated by this, perhaps one can guess that there is a new useful concept to be invented/discovered when one notices that one's graph of allowed inferences has a complete bipartite subgraph which is not already explained by a concept.
 - Suppose one has a bunch of interesting properties $A_1, A_2, \dots, A_n$ with a bunch of the nm pairwise inferences to the interesting properties $B_1, B_2, \dots, B_m$ already considered valid.
 - Then the hypothesis says one should investigate if these inferences can be "factored through" a new property X like $A_i \Rightarrow X \Rightarrow B_j$ (where by an implication between properties I mean that an object having one property implies the object having the other property as well).
 - If one really has all the pairwise inferences $A_i \Rightarrow B_j$ already considered valid, then one can always introduce the "artificial" new concept $X = A_1 \text{ OR } \dots \text{ OR } A_n$ or $X = B_1 \text{ AND } \dots \text{ AND } B_m$ (either choice satisfies the property required above). But I think one would like to have a new concept X that is simple and/or easy to work with. In particular, we want it to be easy to prove each $A_i \Rightarrow X$ and $X \Rightarrow B_j$.
- [See here for more on this.](#)

Open problem in understanding concepts: parsing music

- When listening to a song, a human receives a pressure time series as an input and understands it in terms of the following:
 - the pitch and loudness
 - notes
 - the melody, different voices
 - chords, chord sequences, the harmony
 - the scale
 - the rhythm, the tempo, bars
 - the chorus and verses
 - instruments that are playing
 - the genre
 - phonemes, words, sentences, noticing that some things rhyme, the story being told
 - the artist(s), composer, lyricist
 - tons of more fancy music ideas you learn to think in terms of in eg composition school
- Intuitively, it feels like it should be possible for an AI to start only with a big data set of human music (presented as a bunch of pressure sequences) and invent many of these notions from scratch.
 - We'd e.g. certainly want the AI to come up with an idea of pitch/period and notice that there are long sequences where the period is constant. Hopefully it performs sth like a fourier decomposition into sine waves and notices that each fourier component persists for a while.
 - Other things are probably really hard to invent with only this data, e.g. understanding meanings of words is harder. Knowing certain things about artists will be very hard.
- You can think about how you would go about doing this if you couldn't hear music, and instead needed to look at pressure time series data.
- It'd also be interesting to have a genealogy of these sonic concepts in the human case.
 - Like, how come we ended up with all these concepts? One should be able to tell a developmental story.
- You can ask the same question with a data set of structures on a 2D domain: pictures.

Some thoughts on how to solve this with a verifier picture

- I like the idea of adding more verifiers in stages, with the first-layer of verifiers having to relate to raw data, but with next layers being allowed to relate only to other verifiers.
- I like the idea of asking for predictions of specific aspects of the data, not necessarily the raw data. I.e., you can define new readoffs on the raw data, with predictable/regular values.
 - E.g. you can have a verifier that looks for a pitched component, and a law that predicts these stay present for a while.
- Unfortunately this has trivial solutions. There needs to be some force pushing the verifiers to be interesting. We could say that they should serve some purposes (eg maybe doing a fourier decomposition is good for recognizing objects by sound? at least, there are good reasons to think that object recognition should only care about the fourier decomposition and [not the relative phases](#)), and maybe that's even correct in practice, but that doesn't help us here.
- See [here](#) for some messy notes on this problem.

Concluding thoughts on coming up with good models

- I don't know how producing good models works.
 - Some obstacles:
 - I don't really know what models are, or if they are a natural thing at all.
 - I don't have a clear idea of how to tell if a model is good.
 - My guess is that coming up with good models is a really complicated thing, "an art".
 - But there is surely a lot to be understood about how humans come up with good models and much more still to be understood about how one could come up with good models.
- One idea is that models are built of concepts, and maybe we can understand how to come up with good concepts.
 - Probably good concepts are concepts that facilitate the creation of good models.
 - But more concretely, maybe good concepts are ones that make useful inferences easy?
 - Coming up with good concepts is probably a really complicated thing, "an art".
 - A reason for this is that coming up with the right concepts is a central component of coming up with good models and proofs.
- I haven't really even said what type of thing a concept is!

Thank you for listening!

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